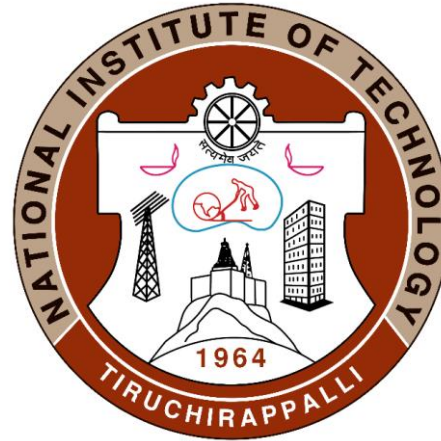


Introduction to Polymers as an Engineering Material



Dr. Nimu Chand Reger (Assistant Professor)

Metallurgical and Materials Engineering

National Institute of Technology Tiruchirappalli, Tamilnadu 620015

nimu@nitt.edu

Lecture - 4

Polymers for aerospace application

Outline

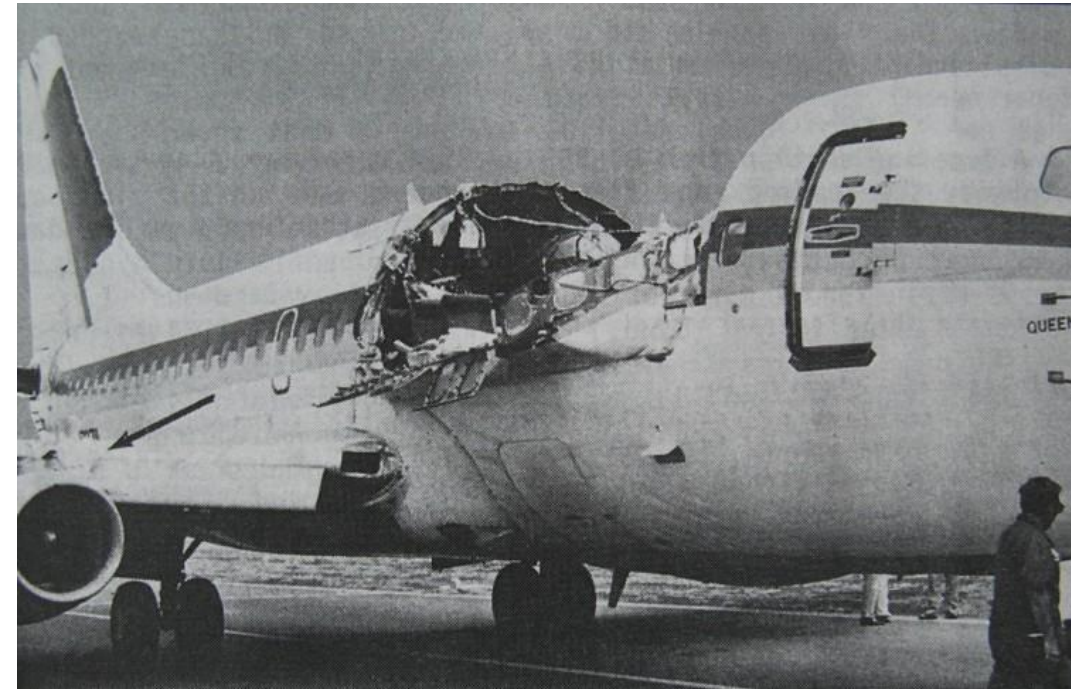
- ❑ Introduction
- ❑ Roles of polymers
- ❑ Applications of polymers
- ❑ Recent advancement

Why Polymers in Aerospace Industry?

☐ Safety

☐ Weight Reduction

☐ Corrosion Damage



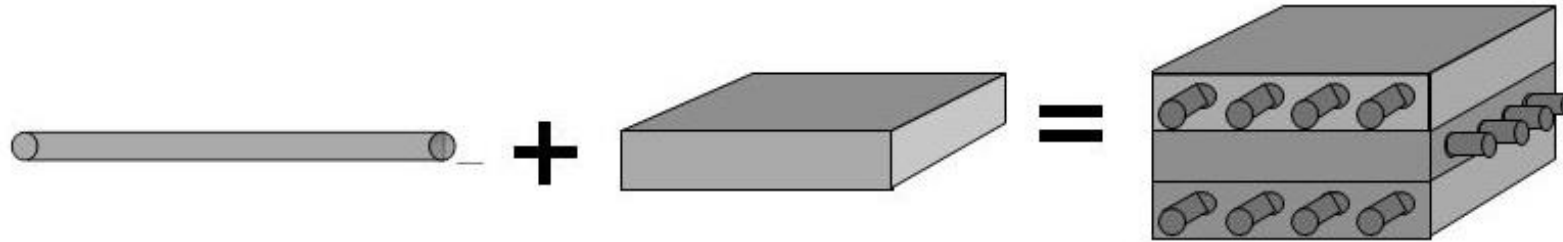
History of Aerospace Materials

- ❑ Early aircraft design used combination of Wood and high tension wires
- ❑ Carbon fiber and glass fiber reinforced polymer first included in aircraft during 1970s.
- ❑ Nowadays most of aircraft parts are made up of polymers because of their useful properties

Polymer Composite in Aerospace

- What is composite?
 - ❑ **Composite** material is a material composed of two or more distinct phases which are chemically and physically different.
 - ❑ The primary phase, called **Matrix** phase is more ductile and less hard phase.
 - ❑ The second phase which is embedded in matrix is called **Reinforcing** phase which is stronger than matrix

Composition of Composites



Fiber/Filament Reinforcement

- High strength
- High stiffness
- Low density
- Carbon, Glass, Aramid, etc

Matrix

- Good shear properties
- Low density
- Thermoset & Thermoplastic
- Epoxy, Polyester, PP, Nylon, Ceramics, etc.

Composite

- High strength
- High stiffness
- Good shear properties
- Low density
- **Anisotropic!**

Polymer Matrix

☐ Polyphenylene Sulphide (PPS)

☐ Polyether ether Ketone(PEEK)

☐ Polyetherimide (PEI)

☐ Polyester

Properties Polymer Composites

- ❑ High Strength to weight ratio
- ❑ High Creep Resistance
- ❑ Lower Thermal Expansion Properties
- ❑ Good Dimensional Stability

Applications

☐ Fuselage

☐ Wings

☐ Tail part



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AERO



Aircraft Fuselage



ROLE OF POLYMERS IN AEROSPACE

- Composites
- Coatings
- Elastomers
- Adhesives

ADHESIVES

- Used for joining the different materials
- **Adhesion bond** – It is the Bonding of one material to the other

ADHESION



COHESION



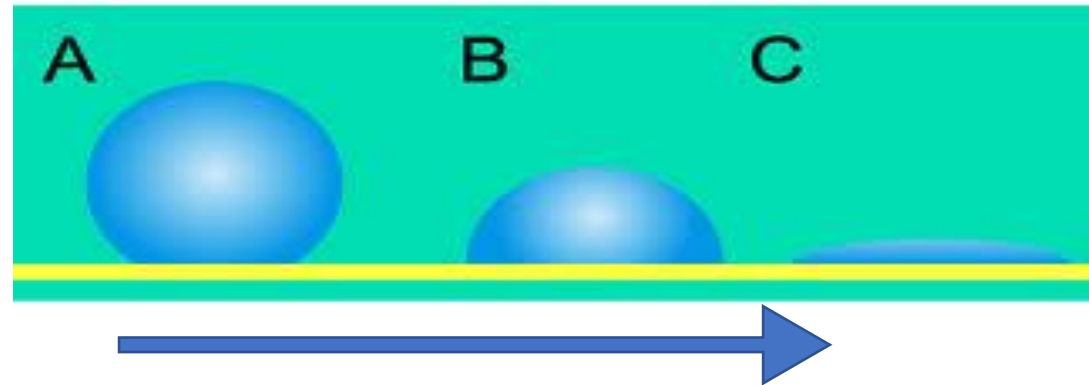
EXAMPLES

Epoxies, polyurethane, cyanoacrylates

Adhesion bonding : Novel joining process for aerospace to join different materials with the help of polymer adhesives.

Basics of adhesion

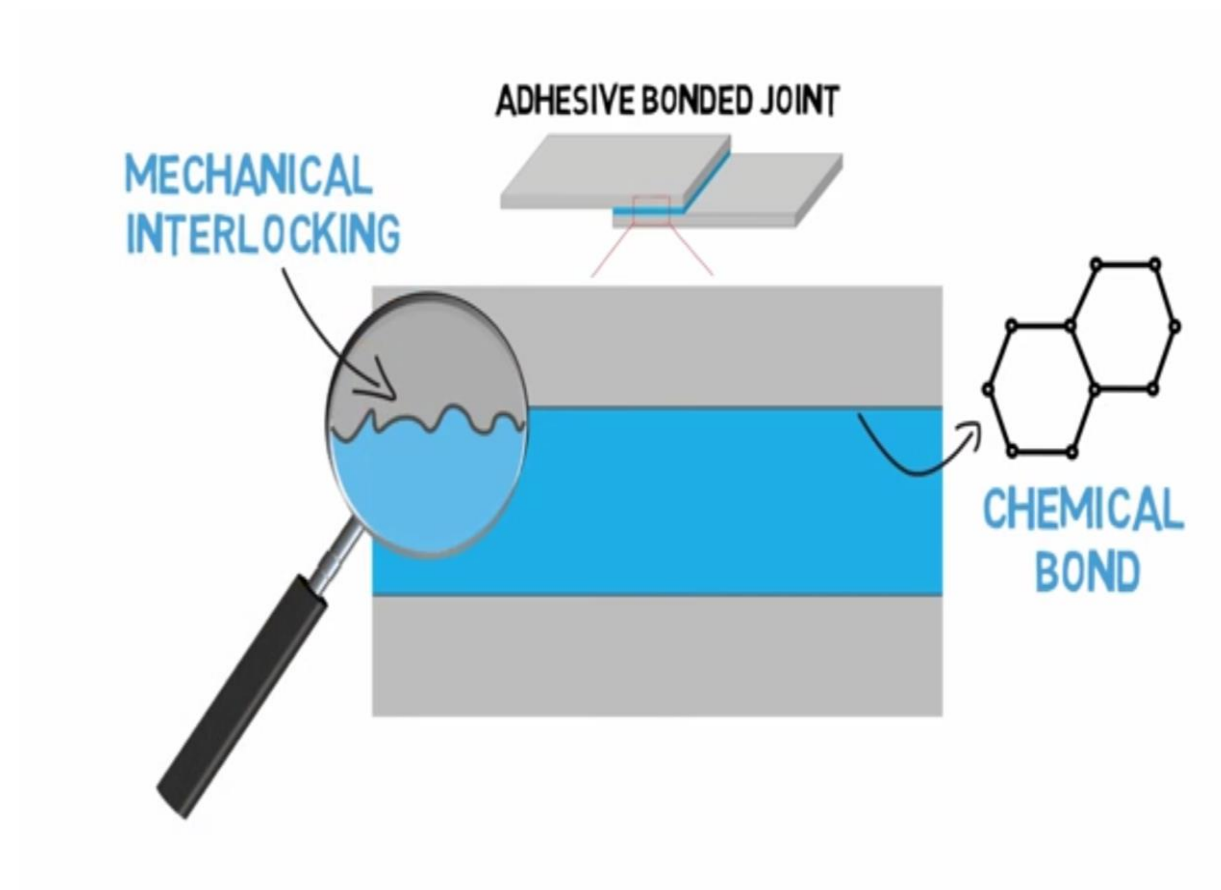
poor
wetting



excellent
wetting

Wetting : Ability of an adhesive to maintain effective contact with the surfaces.

Methods of adhesion



Basic steps

- **Degreasing**

To remove

Grease,oils,dust,rust

Two methods

Wiping the surface

Vapour degreasing

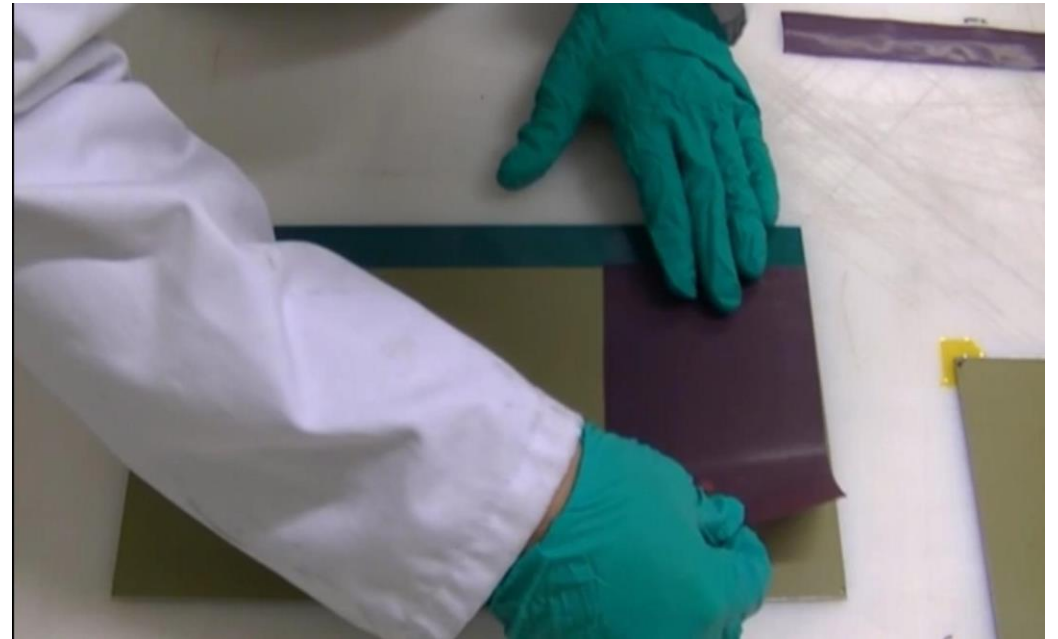
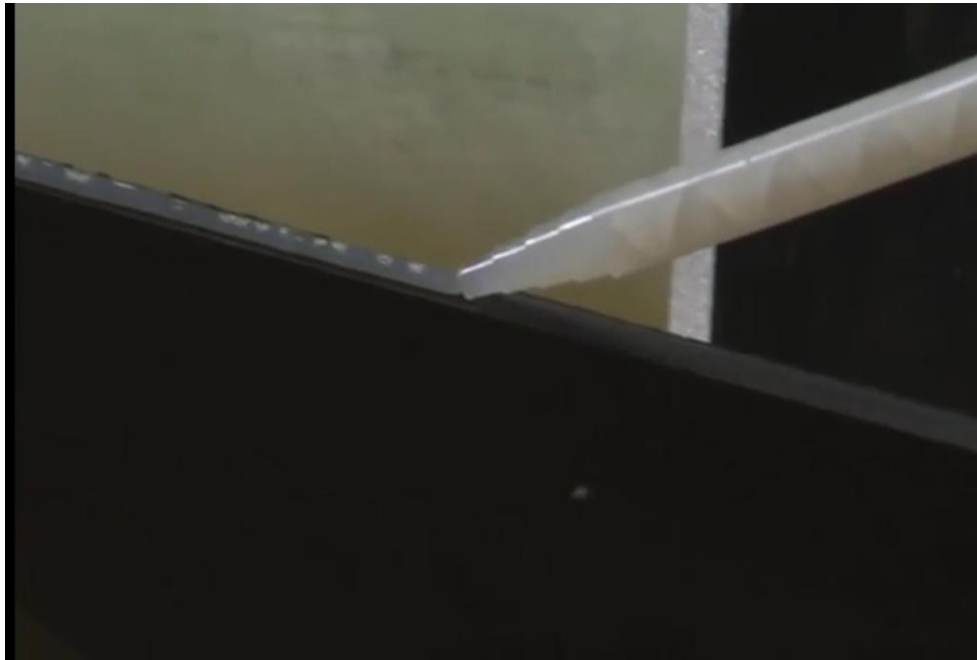


Abrasion



Adhesive application

- Adhesives hold the surface together



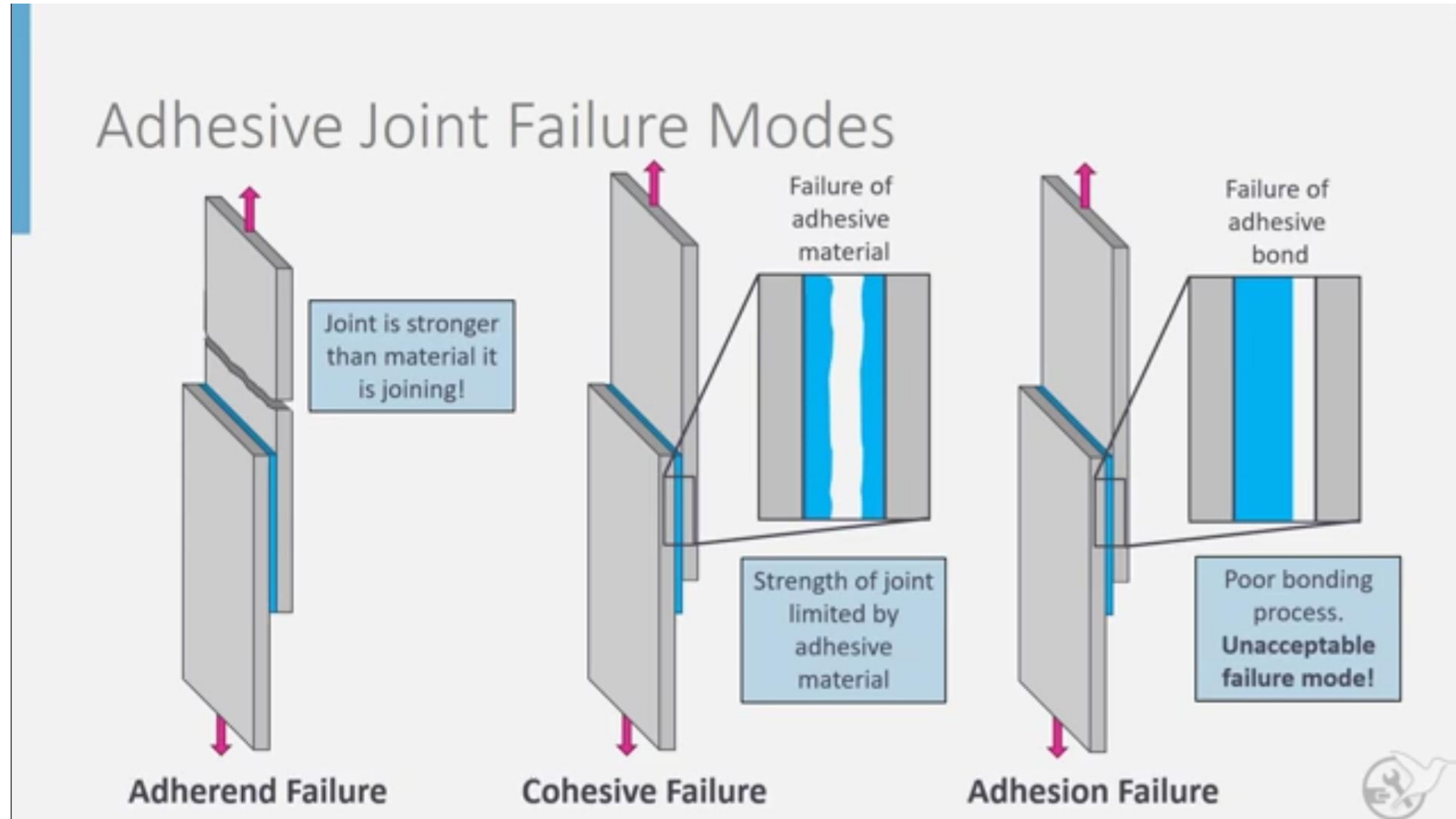
Curing

- Heating is done to speed up the process which accelerates the chemical reactions within adhesives



Curing Process

Modes of failure

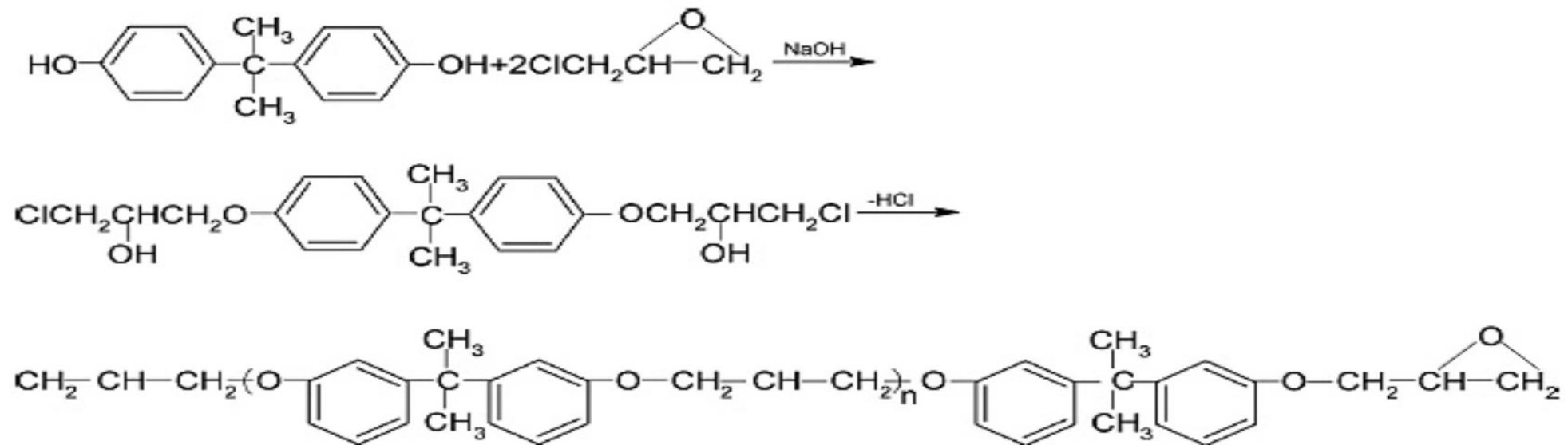
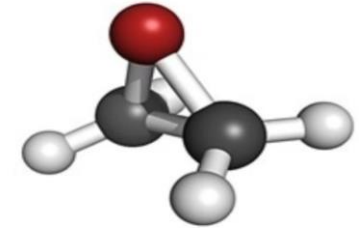


Types of polymer adhesives

- **Epoxies**
- **Polyurethanes**
- **Cyanoacrylates**
- **Modified phenolics**
- **Hot melts**
- **High temperature adhesives**
- **Rubber adhesives**

Epoxies

- Characterised by the presence of an **epoxide group**
- Synthesis of diglycidyl ether of bisphenol A (DGEBA), the most widely used epoxy resin monomer



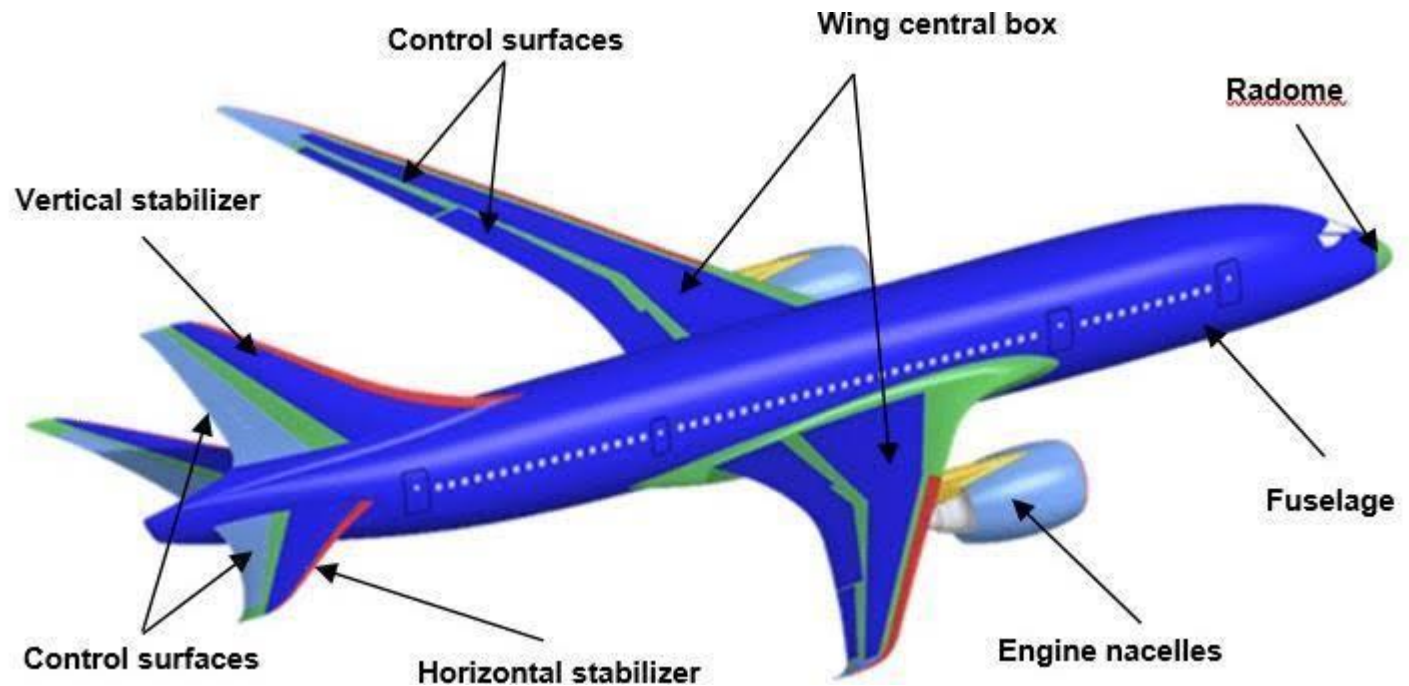
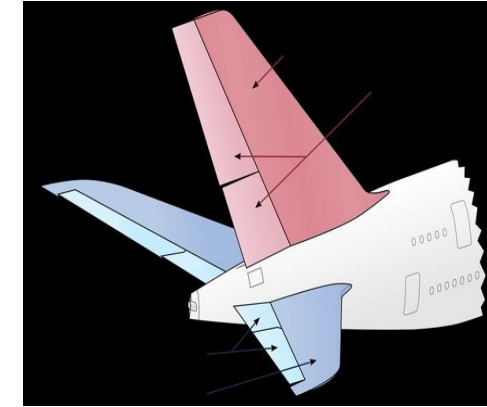
Synthesis of Epoxy Monomer from Bisphenol A and Epichlorohydrin

Properties

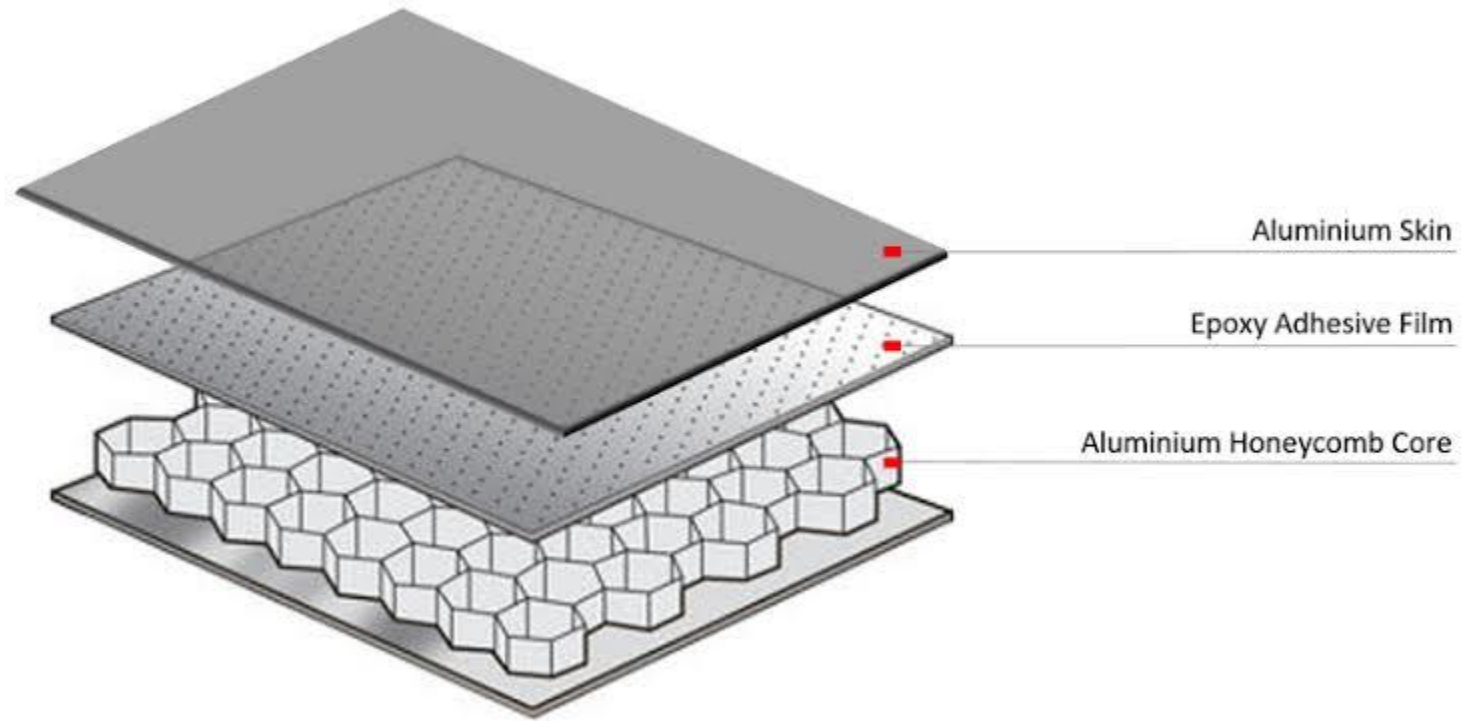
- **High strength**
- **Low shrinkage**
- **Excellent adhesion to various substrates**
- **Low toxicity and low cost**
- **Resistant to wear, cracking, corrosion damage**

Applications

- Fuselage
- Cabin
- Empennage
- Radomes



Adhesive application



Advantages

Join different substrates

Uniform stress distribution

Good damping properties

Disadvantages

Limited stability at higher temperatures

Surface must be cleaned

Limited NDT

Plastics in the Aerospace Industry

- Plastics are found in commercial planes, military aircraft and even satellites. While each plastic has unique purposes, it is also crucial to remember not all plastics are suitable for use in this field. High-performance plastics satisfying various standards are required to maximize performance and assure safety.
- A wide range of parts in the aerospace industry are made from plastics
- **Here are some common applications of plastics in the aerospace industry:**
 1. Aerostructure: wings, doors, pipes, leads, cable ducts
 2. Components: fasteners, bearings, bushings, refueling and fuel systems
 3. Cabin interior: seating, cabin lighting, drinking water systems
 4. Satellites: antenna covers, bearing bushes, sliding elements
 5. Propulsion systems: engines and components

Most common plastics used in the aerospace industry

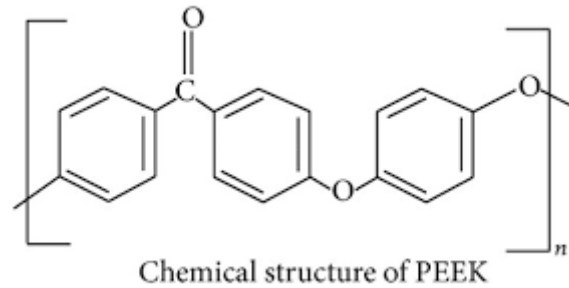
❑ Poly ether ether keton (PEEK):

PROPERTIES:

- High creep resistance and low flammability
- **PEEK** can also withstand large doses of radiation
- It has excellent hydrolysis resistance
- Broad temperature range
- Low fire, smoke and toxic emissions

❑ Applications:

1. Clamps and stand-offs
2. Structural brackets
3. Thermal acoustic insulation
4. Tubing systems
5. Fasteners and connectors



Thermosetting polyimide

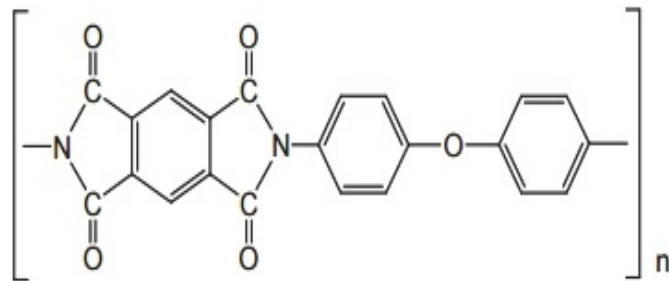
- Properties:

1. Excellent mechanical properties and high resistance to chemicals
2. Higher ductility than ceramics and lower weight than metal alloys

- Applications:

1. Electrical stand offs (spacers)
2. Insulators for threaded nuts and other components

- Structure



Poly chloro trifluoro ethylene (PCTFE)

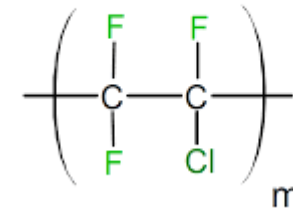
- Properties:

1. Dimensionally stable, rigid and resistant to cold flow
2. Very low gas permeation and outgassing
3. Nearly zero moisture absorption
4. Excellent chemical resistance
5. Useful temperature range -400 F to 380 F
6. Radiation resistance

Applications :

- Cryogenic and chemical processing components
- Seals and gaskets
- Aerospace valve seats, pump parts, impeller diaphragms and plugs
- Liquid oxygen and liquid nitrogen valve linings

Polychlorotrifluoroethylene, PCTFE



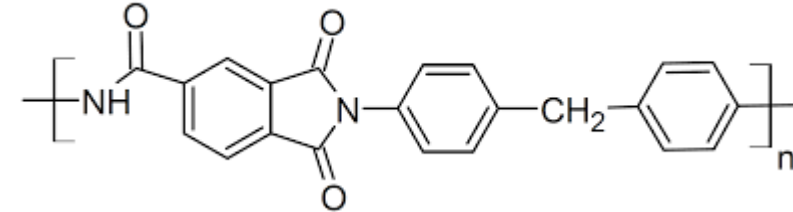
Polyamide-imide(PAI)-Torlon

Properties:

- High strength
- Outstanding creep and wear resistance
- Self lubrication
- Low coefficient of hygroscopic expansion and thermal expansion
- Flame retardant

Applications :

- ☐ Bearings
- ☐ Gears
- ☐ Fastening screws
- ☐ Thermal insulators
- ☐ Air and fuel connectors
- ☐ Compressor valves and seals



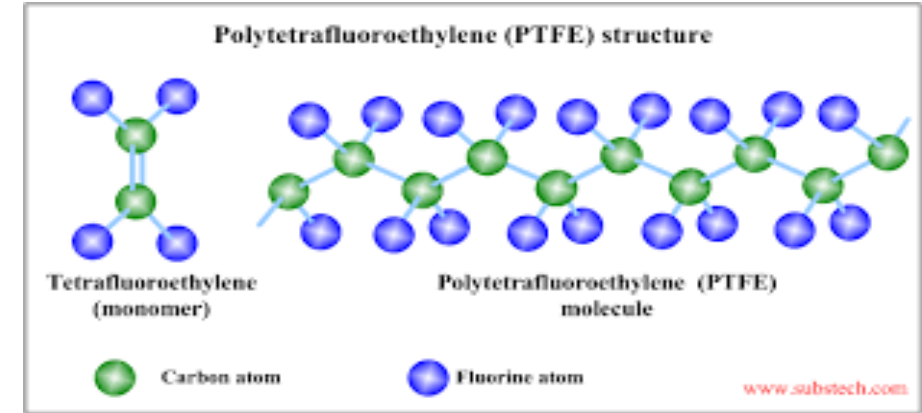
Poly tetra fluoro ethylene(PTFE)-Teflon

Properties:

1. Corrosion resistance
2. Electrical insulator
3. Extremely low coefficient of friction
4. Inflammability
5. Retain physical properties at temperature between -195C to 260C

Applications:

- Aircraft wings
- Electrical connectors
- Fuel systems



Recent Advancements of Polymers in Aerospace Industry

- Development of Self-Healing Coatings in the Aerospace industry.
- Hydrophobic and Oleophobic Thin film coatings.

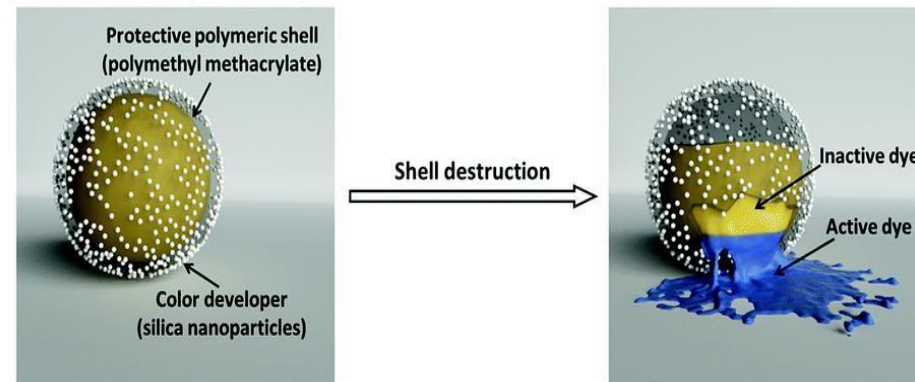
Self-Healing Coatings in Aerospace Industry

- As the name suggests, self healing materials repair damage to themselves without human intervention.
- Self healing systems are split into intrinsic and extrinsic methods.
- The extrinsic self-healing method follows two main design outlines.
- The hollow tube or the use of vascular systems
- The capsule based system

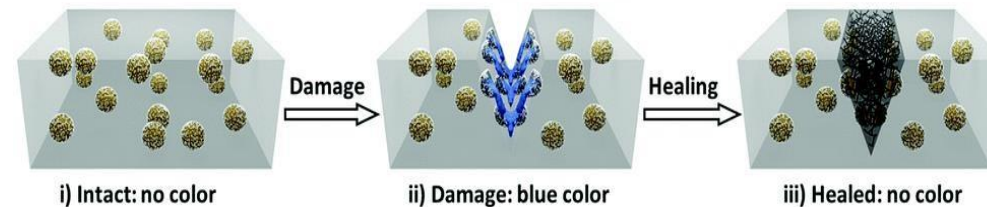
Capsule based system

- Microcapsules based systems use the healing agents in discrete capsules
- The healing agent is encapsulated in a substance that ruptures under a specified threshold when damage occurs.
- Once a capsule is broken, the previously encapsulated monomer mix with the catalyst, leading to polymerization reaction without external heat, and the crack is healed.

a) Damage self-reporting capsules



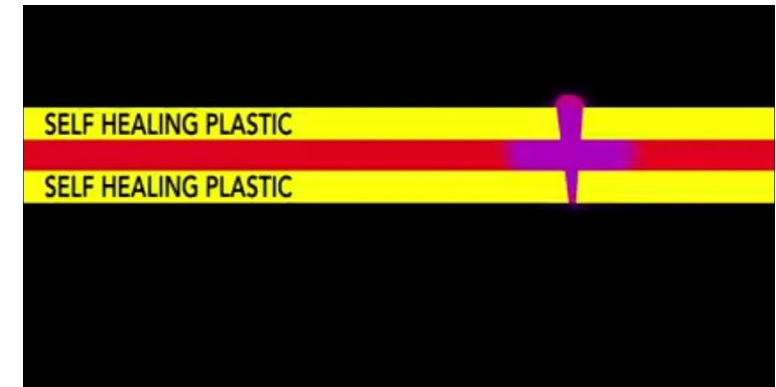
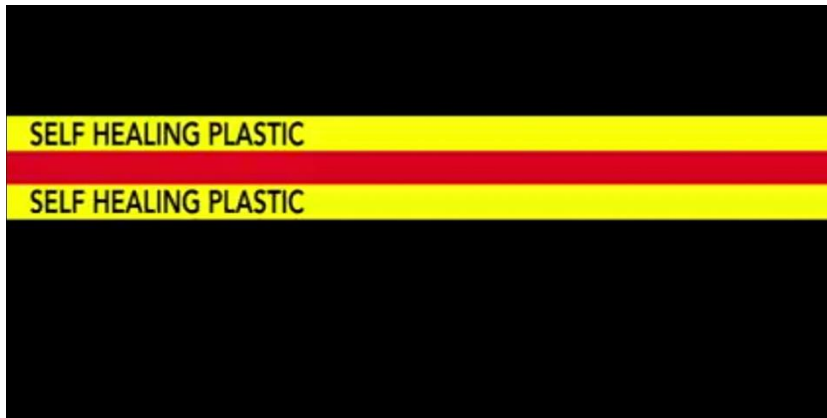
b) Indication of crack appearance and healing in coatings



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Hallow tube system

- Tributylborane is a gel stored between two thin sheets and when a hole is made in one of those sheets the gel moves to fill the hole ‘
- The gel then hardens upon contact with oxygen



Hydrophobic and oleophobic thin film coatings needs

- ✓ Fogging and moisture buildup negatively impact optical performance
- ✓ Cleaning optical surface is time consuming ,expensive or impossible
- ✓ Finger prints negatively impact performance of optics
- ✓ Contamination and fouling materials negatively impact optical behavior
- ✓ Improved adhesive /bonding characteristics are desired on optical surfaces

Hydrophobic and oleophobic thin film coatings

- Aculon's thin-films coatings create a water resistant coating (hydrophobic) and oil-repellent (oleophobic) surfaces to repel water, dust, oil, and dirt on optics.
- Aculon's non-stick coatings, pro-stick/adhesion coatings, and anti-oxidation coatings can be used on a broad range of materials used in industrial applications including metals, metal oxides, polymers, and ceramics.
- Aculon has developed cleaning solutions that not only do a great job of cleaning stainless steel, jewellery, and automotive wheels, but coat the surfaces as well.

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